INTERNSHIP REPORT

ON

Automation of Coursera Web Application To Identify Courses

Yeshwantrao Chavan College of Engineering

Bachelor of Technology in Computer Science and Engineering (IoT)

Submitted by:

Payal SamarKumar Sahu

Supervised by:

Dr. Piyush Ingole	Ashly M. Benny
(Faculty Supervised)	**(Industry Supervisor)**

Nagar Yuwak Shikshan Sanstha's

YESHWANTRAO CHAVAN COLLEGE OF ENGINEERING,

(An Automation Institution Affiliated to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur)

NAGPUR – 441 110

2025-26

CERTIFICATE OF APPROVAL

Certified that the Internship project entitled **“Automation of Coursera Web Application to Identify Courses”** has been successfully completed by **Payal SamarKumar Sahu** during session 2025-26.

Faculty Supervisor

Signature :

Name : Dr. Piyush Ingole

Designation : Assistant Professor

Industry Supervisor

Signature :

Name : Ashly M. Benny

Designation : Contractor, CPO L&D GenC Training

ACKNOWLEDGEMENT

I, **Payal SamarKumar Sahu** would like to convey my gratitude to Yeshwantrao Chavan College of Engineering **Ashly M. Benny, Contractor CPO L&D GenC Training, Cognizant Technology Solution** for emphasizing on the Semester Internship Program and giving me the platform to interact with industry professionals.

I would also like to thank **Dr. Piyush Ingole** and **Dr. Kavita Singh** for giving me the opportunity to work on the prestigious Internship

I extend my warm gratitude and regards to everyone who helped me during my internship.

13-Jan-2026

Candidate ID: 39901925

Payal Samarkumar Sahu
B.Tech CSE (IOT)
Yeshwantrao Chavan College of Engineering, Nagpur

Dear **Payal Samarkumar Sahu,**

Further to our Letter of Intent for the position of Programmer Analyst Trainee / Programmer Analyst aligned to the hiring category, we are pleased to offer you an internship with us at Cognizant office for **a period of 3 to 6 months**. Your internship on-boarding will be scheduled based on your availability, factoring your college exam schedule, availability of your Provisional Certificate and our business requirements.

During this period, you will be provided with a stipend of **INR 12,000/-** equated to the planned duration of the Internship curriculum and will be paid monthly based on your attendance, subject to eligibility for the monthly payroll processing for a given month, except for the last month's stipend, which will be paid upon successful completion of the learning curriculum.

Though Cognizant Internship is a skilling program aimed at enhancing technical acumen, it does not guarantee employment and there is no employer – employee relationship during the course of this internship program. However, the successful completion of internship may form a critical part of your eligibility for employment with Cognizant if an opportunity arises in future.

You will be provided a learning curriculum as per the skill track assigned to you. The learning design would expect you to drive your learning through hands on exercise and project work. There will also be a series of webinars, quizzes, Subject Matter Expertise (SME) interactions, mentor connects, code challenges, assessments etc. to accelerate your learning. The outcome during Internship would be monitored through formal evaluations.

Prior to joining on the rolls of Cognizant, if offered, you must have successfully completed the prescribed Internship program. In the event of unsatisfactory performance during the Internship or non-completion of the Internship, no Internship Completion Certificate shall be issued by Cognizant. Cognizant reserves rights at its sole discretion to revoke its Letter of Intent.

Section A: Terms and Conditions:

1. The Internship timings would be for 10 hours per day from Monday through Friday, aligned with the working timings followed in Cognizant, which, based on the need, could also be operated on a shift model. Attendance is mandatory on all the days to stay active in the Internship Program. The Internship Offer would be cancelled if the mandatory requirement of minimum 85% attendance at office is not met in a month.

2. Interns are covered under Cognizant's calendar holidays of the respective location of internship, and you would need to adhere with minimum attendance requirements. Prior approvals are must towards any unavoidable leave or break requests during the program and the internship would be cancelled if leaves are availed without prior approvals.

3. You would be required to ensure timely completion and submission of assignments, project work and preparation required prior to the sessions, failing which your internship would be cancelled.

4. The Technical skills track mapped could change at the start or mid-way or even later during the program, depending on business demand changes and you would be required to be flexible for this change, failing which your internship would be cancelled.

5. Stipend payment will be done for the prescribed Internship Curriculum period only and no additional payment will be done for any delay in completion. Minimum 85% of attendance is mandatory for the successful processing of your monthly stipend payment.

6. There would be zero tolerance to plagiarisms and misconduct during the internship. Adherence to Cognizant Internship policies and guidelines is mandatory and any breach or incident reported will lead to immediate cancellation of Internship without any notice. You would be required to complete Cognizant mandatory trainings such as Code of Conduct and Acceptable Use Policy (AUP) within the given timelines.

7. During the course of your Internship and at all times, you shall be governed by Cognizant's Social Media Policy and shall, refrain from posting malicious, libelous, defamatory, false, obscene, political, anti-social, abusive, and threatening messages/statements or disparaging the Company, clients, associates, competitors, or suppliers or any third parties, irrespective of whether any such statements are likely to cause damage to any such entity or person. Any breach of this section would lead to immediate cancellation of the Internship.

8. Cognizant reserves rights regarding IT infra as applicable and access to information and materials of Cognizant during the internship period and may modify or amend the Cognizant GenC program terms and conditions from time to time.

9. It is hereby clarified that participation in this Internship shall not constitute you to be an employee of Cognizant nor shall it obligate Cognizant for any purpose whatsoever. The scope of this Internship does not include any supervisory responsibilities and that there is no agency, fiduciary or employer-employee relationship intended or created by reason of this document.

10. Cognizant holds all rights to cancel this Internship Offer owing to any reason at any point in time during your Internship, including due to non-conformance of performance benchmark or moral code of conduct or in case of you failing to participate in the Internship within the given date/timeline or for any other reasons upon providing written communication of the same to you. Upon such cancellation of this Internship Offer, your access and participation in the Internship shall stand cancelled.

11. At the time of your reporting for the internship, you will be required to sign a Non – Disclosure Agreement with the Company. During the course of your internship and after completion of the same, you are required to maintain strictest confidentiality with respect to Company proprietary or products that you access or come into contact with, during your project as an Intern, at all times as per our Policy. Use of company proprietary information or products shall not be made without prior permission from the concerned authority. Any breach of information security will be dealt as per Company Policy.

12. After successful completion of your internship if there is a business demand which expects you to get enabled on a different skill, you would be onboarded as full-time employee (FTE) and deployed into another formal training based on business demand to a specific skill track which will be used as basis towards your allocation to projects/roles. Successful completion of this training is mandatory to continue as an FTE with Cognizant.

13. This offer from Cognizant shall be active and **valid for only 3 calendar days** and hence you are expected to accept or decline the offer through the Company's online portal within the said time-period of 3 calendar days and you will also be required to submit the mandatory documents at least 10 **days** before your Internship Onboarding Date as part of your Pre-joining & Background Verification (BGV) process. In case you do not comply to the above timelines or if the background verification checks reveal unfavorable results at any time, this Offer shall stand withdrawn and will be considered as cancelled. Any official written extension to the offer validity and the above-mentioned timelines will be at the sole discretion of Cognizant.

14. For avoidance of doubt, it is herewith stated that the Internship shall stand cancelled on the below scenarios as well:

a. In the event of you accepting this Internship Offer but not joining into the Internship on the specified date

and at the specified location of on-boarding.
b. In the event of you not accepting this Internship Offer or failing to communicate acceptance within 3 calendar days as stated above, the opportunity to do internship will be cancelled.
c. For such other operational, regulatory reasons including breach of terms herein.

Thereupon, your access shall also stand revoked, and Cognizant shall not be obligated to extend nor be liable for any claims due to cancellation of this Internship Offer.

On the occurrence of any of the above-mentioned scenarios **(Refer to Section A: Terms and Conditions)** , Cognizant reserves the right to revoke your Internship Offer at its sole discretion. Upon cancellation of your Internship Offer your Letter of Intent would also be revoked at the sole discretion of Cognizant.

Below are the **mandatory documents** to be submitted as part of your **Background Verification:**
• Your Pan Card

Below are the **mandatory documents** to be submitted as part of your **Pre- joining formalities:**
• 2 Passport sized Photographs preferably with a White background
• Personal individual bank account from a nationalized bank for processing stipend

In case of additional queries or concerns, you can raise a query at
https://campus2cognizant.cognizant.com

We wish you good luck.

Yours sincerely,
For **Cognizant Technology Solutions India Pvt. Ltd.,**

Atul Sahgal
Senior Vice President - Talent Acquisition

I have read the offer, understood and accept the above mentioned terms and conditions.

Signature: **Date:**

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INTRODUCTION OF THE REPORT

1.1 Background

The rapid growth of digital technology has transformed the way education and professional training are delivered across the globe. Online learning platforms such as Coursera have become integral to skill development, offering thousands of courses across diverse domains including programming, data science, language learning, and business management. As these platforms grow in scale and complexity, ensuring their functionality, reliability, and user experience becomes increasingly important.

In the software industry, Quality Assurance (QA) plays a vital role in ensuring that web applications perform as expected under various conditions. Manual testing, while essential, is time-consuming, repetitive, and prone to human error. Web automation testing addresses these limitations by allowing testers to simulate real user interactions programmatically, execute test scenarios consistently, and extract data efficiently from live web applications.

This internship was undertaken at Cognizant Technology Solutions, where the student was exposed to real-world automation testing practices. The project focused on automating key user workflows on the Coursera platform using browser automation tools, thereby gaining practical experience in the field of software testing and web automation.

1.2 Purpose of the Report

This report has been prepared to fulfil the requirements of the Semester Internship Program as prescribed by Yeshwantrao Chavan College of Engineering, Nagpur, under the Bachelor of Technology in Computer Technology program for the academic year 2025-26.

The report serves the following specific purposes:

- To provide a comprehensive and structured account of the internship experience undertaken at Cognizant Technology Solutions under the guidance of the industry and faculty supervisors.

- To document the problem statement, objectives, methodology, tasks performed, observations, and outcomes of the automation project carried out on the Coursera web application.
- To demonstrate the practical application of technical knowledge and skills acquired during the B.Tech program in a professional industry environment.
- To reflect upon the learning outcomes, challenges faced, and competencies developed during the internship period.

1.3 Objective of the Internship

The following objectives were defined at the commencement of the internship:

- To understand and apply web automation testing concepts in a real-world industry setting under professional supervision.
- To design and implement automation scripts for identifying and extracting course-related data from the Coursera platform.
- To handle dynamic web elements including search bars, dropdown menus, multi-window navigation, and web forms using browser automation techniques.
- To perform form validation testing by submitting incorrect input data and programmatically capturing the resulting error messages.
- To develop proficiency in industry-standard automation tools and frameworks, and to apply them effectively to solve practical QA problems.
- To enhance professional skills including documentation, problem-solving, analytical thinking, and collaboration within a team environment.

1.4 Scope of Work

The internship project was scoped around three specific automation tasks to be performed on the Coursera website (coursera.org):

Task 1 – Web Development Course Identification: To search for beginner-level web development courses offered in the English language on Coursera, and to extract and display the course name, total learning hours, and rating for the first two courses appearing in the search results.

Task 2 – Language Learning Data Extraction: To navigate to the Language Learning category on Coursera, extract all available languages and their corresponding proficiency levels, and display the total number of courses available under each language and level combination.

Task 3 – Form Validation and Error Capture: To navigate from the Coursera homepage to the "For Enterprise" section, access the "Courses for Campus" product page under the Products menu, fill the "Ready to Transform" contact form with one intentionally invalid input (such as an incorrectly formatted email address), and programmatically capture and display the error or warning message generated by the form validation.

1.5 Tools and Technologies Used

Tool/Technology	**Purpose**
Selenium WebDriver	Browser automation and web element interaction
BDD(Behaviour Driven Development)	Writing test scenarios in plain English using Gherkin syntax (Given/When/Then)
TestNG	Test execution framework for managing and running automated test cases
ExtentReports	Generating detailed HTML test execution reports with logs and results
GitHub	Version control and source code management
Java	Primary programming language for automation scripting
ChromeDriver	WebDriver for Google Chrome browser
Coursera.org	Target web application for testing
IntelliJ IDE	Development and script execution environment

1.6 Methodology

The internship project was executed following a structured and systematic approach consisting of the following phases:

Phase 1 – Requirement Analysis: The problem statement was carefully analyzed to identify the web elements, pages, and interactions involved in each of the three automation tasks.

Phase 2 – Environment Setup: The required software tools, browser drivers, and automation frameworks were installed and configured on the development system to prepare the working environment.

Phase 3 – Script Development: Automation scripts were written to perform each task as per the defined scope, incorporating logic for element identification, data extraction, form interaction, and error capture.

Phase 4 – Execution and Observation: The developed scripts were executed on the live Coursera website. Outputs were observed, verified, and recorded for each task.

Phase 5 – Documentation: All findings, observations, and outcomes were systematically documented and compiled into this internship report as per the format prescribed by Yeshwantrao Chavan College of Engineering.

OVERVIEW OF THE INTERNSHIP PROVIDING ORGANIZATION

2.1 About Cognizant Technology Solutions

Cognizant Technology Solutions is one of the world's leading professional services companies, helping clients transform their business, operating, and technology models for the digital era. Headquartered in Teaneck, New Jersey, USA, Cognizant operates across multiple continents and serves clients in industries including banking and financial services, healthcare, retail, manufacturing, and technology. With a global workforce of over 300,000 professionals, Cognizant combines a passion for client satisfaction, technology innovation, deep industry expertise, and a community-minded approach to drive digital transformation.

Cognizant is listed on the NASDAQ stock exchange under the ticker symbol CTSH and is consistently ranked among the most admired companies in the IT services industry. The company has earned recognition for its culture of continuous learning, its investment in talent development, and its commitment to delivering high-quality technology solutions to clients across the globe.

2.2 Quality Engineering and Assurance (QEA) – Service Line

The internship was undertaken under the Quality Engineering and Assurance (QEA) service line of Cognizant, which is a dedicated division focused on ensuring the quality, reliability, and performance of enterprise applications, processes, and systems.

Cognizant's QEA service line offers a comprehensive range of services including intelligent and automated quality assurance, modernization assurance, and experience assurance. These services are designed to accelerate business and technology changes, improve customer experiences, and ensure regulatory compliance across various industry domains.

QEA places significant emphasis on automation and Artificial Intelligence to enhance testing efficiency and deliver high-quality outcomes. By leveraging these advanced technologies, the

QEA division helps businesses achieve digital success and maintain a competitive advantage in the rapidly evolving digital landscape. The division has successfully partnered with major enterprises across healthcare, finance, and retail sectors to transform their software quality assurance processes and achieve measurable improvements in software reliability, compliance, and time-to-market.

2.3 Generation Cognizant (GenC) – Internship Program

The internship was part of Cognizant's Generation Cognizant (GenC) program, a structured and comprehensive learning initiative designed to engage young talent with a well-defined technical learning pathway. The GenC program offers interns and fresh graduates the opportunity to interact with Subject Matter Experts (SMEs), understand the corporate environment, and develop the skills required to function as professional software engineers in a real-world setting.

The GenC program is built on the principle of Learner Autonomy, where students take charge of their own learning journey using available tools, resources, and self-paced platforms. The emphasis is more on "learning by doing" than on traditional instructor-led instruction, empowering interns to develop problem-solving skills through hands-on practice and real-world project exposure.

The program followed a Flipped Classroom approach, combining self-paced learning sessions with expert-guided practice sessions. During self-learning hours, interns studied the learning objectives independently through the Udemy platform and the Tekstac coding environment. This was followed by practice sessions conducted by SMEs, who provided guidance, resolved doubts, and helped interns apply the concepts to real-world problem statements.

2.4 Program Structure and Learning Journey

The GenC program followed a structured 65-day learning journey divided into two stages:

Stage 1 – Core Java with SQL covered fundamental programming concepts including Core Java, Object-Oriented Programming, Collections, Exception Handling, File Handling, JDBC, and SQL. This stage laid the foundation required to write efficient and maintainable automation code.

Stage 2 – Functional Testing with Selenium Automation Concepts covered the full spectrum of web automation testing, beginning with Functional Testing fundamentals and progressing through Selenium WebDriver, Page Object Model (POM), Apache POI, TestNG, BDD Cucumber, DevOps tools (Git, Maven, Jenkins, Selenium Grid), API automation using Postman and REST Assured, and Generative AI concepts.

The program concluded with a Business Aligned Project, which was divided into a Mini Project (individual activity) and a Hackathon (group activity). These projects were evaluated by Technical Subject Matter Experts from the Business Unit through Interim and Final Evaluations conducted via video interviews.

2.5 Internship Environment and Supervision

During the internship, the student worked under the guidance of **Ashly M. Benny**, Contractor, CPO L&D GenC Training, Cognizant Technology Solutions, who served as the Industry Supervisor. Academic supervision was provided by **Dr. Piyush Ingole**, Assistant Professor, Department of Computer Technology, Yeshwantrao Chavan College of Engineering, Nagpur, who served as the Faculty Supervisor.

The internship provided daily structured learning, hands-on exercises, code reviews, and project deliverable submissions. Regular mentor interactions and SME-connect sessions ensured that the learning experience was aligned with industry standards and professional best practices.

TRAINING COMPONENT

3.1 Topics Covered

The training was structured across a 65-day learning journey divided into two stages as defined by the Cognizant GenC QEA program:

Stage 1 – Core Java with SQL (Day 1 to Day 21)

- Core Java Fundamentals: Data types, variables, operators, control flow statements
- Object-Oriented Programming: Encapsulation, Inheritance, Polymorphism, Abstraction
- Classes and Objects, Constructors, Keywords (this, super, final, static)
- Collections Framework: ArrayList, LinkedList, HashMap, HashSet, TreeSet, PriorityQueue
- Exception Handling: try-catch, finally, throw, throws, custom exceptions
- Java File Handling: FileReader, FileWriter, BufferedReader, BufferedWriter
- Java 8 Features: Lambda expressions, Stream API, Optional class
- SQL: DDL, DML, Joins, Subqueries, Aggregate Functions, Stored Procedures
- JDBC: Java Database Connectivity, CRUD operations

Stage 2 – Functional Testing with Selenium Automation (Day 22 to Day 65)

- Agile/Scrum Methodology: SDLC, STLC, V-Model, Sprint planning, Retrospectives
- Functional Testing: White Box/Black Box Testing, Smoke, Sanity, Regression Testing
- JIRA: Project creation, Backlog management, Bug logging, Test case management
- Selenium WebDriver: Architecture, locators (XPath, CSS, ID, Name, LinkText)
- Advanced WebDriver: Actions class, JavascriptExecutor, Window/Frame handling, Waits
- Page Object Model (POM): Design pattern, Page Factory, reusable components
- Apache POI: Reading/writing Excel for data-driven testing
- TestNG: Annotations, assertions, XML configuration, parallel execution
- BDD Cucumber: Gherkin syntax, Feature files, Step Definitions, Hooks, Runners
- Automation Framework: Hybrid framework design, Extent Reports, Log4j
- DevOps: Git, GitHub, Maven, Jenkins CI/CD pipeline, Selenium Grid, Docker basics
- API Automation: Postman, REST Assured, JSON/XML parsing, Authentication
- GenAI: Fundamentals of Generative AI, GitHub Copilot, Prompt Engineering

3.2 Technologies and Tools Used

Tool / Technology	Category	Purpose
Java 17 (Azul JDK)	Programming Language	Core language for automation scripting
Selenium WebDriver 4.15.0	Automation Framework	Browser automation and element interaction
BDD Cucumber	Test Design	Behaviour-driven development using Gherkin
TestNG 7.4.0	Test Execution	Test management, grouping, parallel execution
ExtentReports	Reporting	HTML test execution reports with screenshots
Apache POI 5.2.4	Data Management	Reading/writing Excel test data
Maven	Build Tool	Dependency management and build automation
GitHub	Version Control	Source code management and collaboration
Jenkins	CI/CD	Automated test scheduling and pipeline
JIRA	Project Management	Sprint management, defect tracking, test cases
Eclipse IDE 2023-12	Development Environment	Writing, running, and debugging automation code
Postman	API Testing	Manual and automated API testing
REST Assured	API Automation	Java-based REST API automation framework
ChromeDriver	Browser Driver	WebDriver for Google Chrome browser

3.3 Certifications

During the internship period, the following assessments and evaluations were completed as part of the Cognizant GenC program:

- Stage 1 Assessment – Core Java and SQL (Cleared with qualifying score)
- Functional Testing ICT – Integrated Capability Test on Software Testing concepts
- Selenium ICT – Integrated Capability Test on Selenium WebDriver automation
- Interim Project Evaluation + Technical Evaluation – Video interview with BU SME (Green status)
- Final Project Evaluation + Final Technical Evaluation – Video interview with BU SME (Excellent)
- GenAI Fundamentals – Fundamentals of Generative AI [101-Basics] course on C-Learn

3.4 Week-wise Assessment and Scores

Week	Assessment / Activity	Topics Covered	Status
Week 1-2	Core Java Code Challenges (CC) - Group 1 & 2	OOP, Collections, Exception Handling	Cleared
Week 3	ICT - Java Assessment	Core Java comprehensive	Cleared
Week 3	SQL Code Challenges (DDL, DML, Joins, Subqueries)	SQL Queries	Cleared
Week 3	Stage 1 Assessment	Java + SQL combined	Cleared
Week 4	Functional Testing ICT	STLC, Test cases, JIRA	Cleared
Week 5-6	Selenium WebDriver Code Challenges	Locators, XPath, Actions	Cleared
Week 6	Selenium with POM and Apache POI CC	POM, Data-driven testing	Cleared
Week 7	TestNG Code Challenge + ICT	TestNG, XML, JSON	Cleared
Week 7	Interim Project + Technical Evaluation	All Stage 2 topics + Mini Project	Green (Pass)
Week 9	GenAI Knowledge Check Assessment	Generative AI Fundamentals	Cleared
Week 10	Final Project + Technical Evaluation	Complete curriculum + Hackathon	Excellent

ABOUT THE PROJECT

4.1 Project Title

Automation of Coursera Web Application to Identify Courses

4.2 Project Overview

This project was undertaken as part of the Hackathon component of the Cognizant GenC Quality Engineering and Assurance (QEA) internship program. The project involved the design and implementation of a comprehensive web automation testing framework targeting the Coursera platform (coursera.org), one of the world's most widely used online learning portals.

The primary objective of the project was to automate three distinct real-world user workflows on the Coursera website, covering a broad range of automation scenarios including browser-based search and data extraction, dynamic dropdown handling, multi-window navigation, web form interaction, and validation error capture. The project was built using Java as the programming language, Selenium WebDriver for browser automation, BDD Cucumber for behaviour-driven test development, TestNG as the test execution framework, ExtentReports for test reporting, and GitHub for version control. The overall framework was designed following the Page Object Model (POM) design pattern to ensure maintainability, reusability, and scalability of the automation codebase.

4.3 Problem Statement

The project addressed the following problem statement as defined in the Hackathon:

"Search and display all web development courses. The courses should be at beginner level, offered in the English language. Display the first two courses with course name, total learning hours, and rating."

The broader scope of the Hackathon extended this problem statement to include three specific automation tasks:

Task 1: Search for web development courses for beginners level in the English language on Coursera and extract the course name, total learning hours, and rating for the first two courses displayed in the search results.

Task 2: Navigate to the Language Learning category on Coursera, extract all available languages and their different proficiency levels along with the total course count for each, and display the complete extracted data.

Task 3: From the Coursera homepage, navigate to "For Enterprise," access "Courses for Campus" under the Products section, fill in the "Ready to Transform" contact form with one intentionally invalid input such as an incorrectly formatted email address, capture the error or warning message displayed by the form validation, and display it.

4.4 Key Automation Scope

The automation framework built for this project covered the following key technical scenarios:

Handling different browser windows and search functionality was a core requirement, as the Coursera platform dynamically loads content and opens results in the same or new browser contexts depending on the user action. The framework was designed to manage these scenarios using Selenium WebDriver's window handling capabilities.

Extracting multiple dropdown list items and storing them in collections was addressed in Task 2, where the Language Learning section presented multiple dropdown-style filter menus containing language names and proficiency levels. These were extracted programmatically and stored in Java collections such as ArrayLists and HashMaps for structured display and reporting.

Navigating back to the homepage was required between tasks to simulate realistic user journeys. The framework incorporated explicit navigation logic to return to the Coursera homepage using both browser history navigation and direct URL loading.

Filling forms across different web page objects was required in Task 3, where the "Ready to Transform" form contained multiple input fields of different types including text fields, dropdown selectors, and submit buttons, all located within a complex multi-section page layout.

Capturing warning messages was the final objective of Task 3, where the form validation error displayed upon submitting an invalid email address was captured programmatically using Selenium element identification and stored for reporting.

Scrolling down in web pages was necessary throughout the project, as several key elements on the Coursera platform were not visible within the initial viewport and required JavaScript-based or Actions class-based scrolling to bring them into view before interaction.

4.5 Tools and Technologies Used

Tool/Technology	**Purpose**
Selenium WebDriver	Browser automation and web element interaction
BDD(Behaviour Driven Development)	Writing test scenarios in plain English using Gherkin syntax (Given/When/Then)
TestNG	Test execution framework for managing and running automated test cases
ExtentReports	Generating detailed HTML test execution reports with logs and results
GitHub	Version control and source code management
Java	Primary programming language for automation scripting
ChromeDriver	WebDriver for Google Chrome browser
Coursera.org	Target web application for testing
IntelliJ IDE	Development and script execution environment
Apache POI 5.2.4	Reading and writing test data from Excel files for data-driven testing
JIRA	Agile project management, test case documentation, sprint management, and defect tracking
Maven	Build automation and dependency management

Page Object Model (POM)	Design pattern for separating test logic from page element definitions

4.6 Framework Architecture

The automation framework built for this project covered the following key technical scenarios:

Handling different browser windows and search functionality was a core requirement, as the Coursera platform dynamically loads content and opens results in the same or new browser contexts depending on the user action. The framework was designed to manage these scenarios using Selenium WebDriver's window handling capabilities.

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Scrolling down in web pages was necessary throughout the project, as several key elements on the Coursera platform were not visible within the initial viewport and required JavaScript-based or Actions class-based scrolling to bring them into view before interaction.

4.7 Task 1 – Web Development Course Identification

Objective: To search for beginner-level web development courses offered in English on Coursera and extract the name, total learning hours, and rating for the first two courses in the results.

Steps Performed:

The browser was launched and maximized, and the Coursera homepage at coursera.org was loaded. The search bar was located using its XPath locator and the keyword "Web Development" was entered and submitted. On the search results page, the Level filter was applied by selecting "Beginner" from the filter panel on the left side, and the Language filter was set to "English." The page was scrolled down to ensure all course cards were loaded within the viewport. The first two course cards were identified using dynamic XPath locators targeting the course card container elements. For each of the two courses, the course name, total learning hours, and rating were extracted using the getText() method of Selenium WebDriver and stored in a Java ArrayList. The extracted data was then printed to the console and logged into the ExtentReport for documentation.

Challenges Faced: The search results page used dynamic lazy loading, meaning course cards only fully rendered after the page was scrolled. Handling this required the use of JavascriptExecutor for scrolling combined with Explicit Waits using ExpectedConditions to ensure elements were fully visible before data extraction was attempted.

4.8 Task 2 – Language Learning Data Extraction

Objective: To navigate to the Language Learning category on Coursera, extract all available languages and their proficiency levels, and display the total course count for each combination.

Steps Performed:

From the Coursera homepage, the subject catalog was accessed and the Language Learning category was selected. The page loaded a list of language options displayed either as filter dropdowns or category links. Using Selenium's findElements() method with a CSS selector targeting all language filter items, the complete list of available languages was extracted and stored in a Java ArrayList. For each language, the proficiency level options such as Beginner, Intermediate, Advanced, and Mixed were identified from the corresponding dropdown or subfilter panel. The total number of courses available for each language-level combination was extracted from the count label displayed alongside each filter option. All extracted data was stored in a HashMap with the language name as the key and a nested structure of levels and counts as the value, and was then displayed in a formatted output and logged to the ExtentReport.

Challenges Faced: The filter panel containing language and level options was dynamically rendered and required scrolling within the filter container div. Standard scrolling using JavascriptExecutor was used to bring nested filter items into view. Storing multi-level nested data required careful use of Java collections including HashMap and ArrayList to preserve the hierarchical structure of the extracted information.

4.9 Task 3 – Form Validation and Error Message Capture

Objective: To navigate from the Coursera homepage to the "For Enterprise" section, access the "Courses for Campus" product page, fill the "Ready to Transform" contact form with an invalid email input, and capture the resulting error message.

Steps Performed:

The automation script started from the Coursera homepage. The "For Enterprise" option was located in the top navigation bar using a link text locator and clicked. On the For Enterprise landing page, the Products menu was accessed and the "Courses for Campus" option was selected. The script scrolled down the Courses for Campus page until the "Ready to Transform" contact form was visible within the viewport using JavascriptExecutor. The form fields were located using their XPath or CSS ID locators. All mandatory fields such as First Name, Last Name, and Organization were filled with valid test data. The email field was intentionally populated with an incorrectly formatted email address such as "testuser@" to trigger the form's built-in validation. The submit button was clicked, and the script waited for the validation error message to appear using an Explicit Wait. The error message text was captured using the getText() method and stored in a String variable, then printed to the console and logged in the ExtentReport.

Challenges Faced: The "Ready to Transform" form was positioned well below the page fold and required reliable scrolling logic using JavascriptExecutor to bring it into view. The form contained multiple field types including text inputs and dropdown selectors, each requiring different Selenium interaction methods. The error message was rendered as a dynamic DOM element that appeared only after the submit button was clicked, requiring an Explicit Wait with ExpectedConditions.visibilityOfElementLocated to ensure it was fully rendered before the getText() call was executed.

4.10 Test Execution and Reporting

All three automation tasks were executed as part of a unified TestNG test suite configured through a testng.xml file. The tests were grouped into a regression suite and executed sequentially. Upon completion of each test run, ExtentReports automatically generated a detailed HTML report containing the test execution summary, individual step results, pass/fail status, and captured screenshots at key checkpoints.

The test suite was also integrated with Jenkins to enable scheduled and on-demand CI execution. The project source code was maintained on GitHub with regular commits following standard Git workflow practices including branching, committing, and pushing changes. JIRA was used throughout the project for tracking user stories, managing sprint backlogs, logging defects, and maintaining test case traceability.

4.11 Defect Management Using JIRA

All defects identified during test execution were logged in JIRA with the following details captured for each defect: a unique defect ID, the defect summary and description, the steps to reproduce the issue, the expected versus actual result, the severity and priority classification, the environment details, and any screenshots or logs attached as evidence. Sprint reviews and retrospectives were conducted as part of the Agile methodology followed during the project, and all defect statuses were tracked through the JIRA board from Open through In Progress to Resolved and Closed.

WEEKLY WORK LOG

Week	Dates	Tasks Performed	Learning Outcomes	Supervisor Remarks
1	31 Jan - 4 Feb	Corporate induction, Icebreaker sessions, Core Java - OOP, data types, variables, constructors	Understood OOP principles, Java basics and JVM architecture	Good progress
2	7 Feb - 11 Feb	Core Java - Collections, Exception Handling, File Handling, JDBC, hands-on exercises on Tekstac	Implemented ArrayList, HashMap, try-catch blocks, file I/O operations	Satisfactory
3	14 Feb - 18 Feb	Java 8 features, SQL basics, DDL/DML, Joins, Stage 1 Assessment	Mastered lambda expressions, stream API, and SQL query writing	Good
4	21 Feb - 25 Feb	Agile/Scrum, STLC, Functional Testing, JIRA project setup, test case writing	Understood testing lifecycle, created test cases and sprint in JIRA	Good
5	28 Feb - 4 Mar	Selenium WebDriver setup, locators (XPath, CSS, ID), browser automation basics	Wrote first Selenium scripts, identified elements using multiple locator strategies	Excellent
6	7 Mar - 11 Mar	POM design pattern, Page Factory, Apache POI, data-driven testing, advanced XPath	Built first POM framework, read Excel test data, implemented dynamic XPath locators	Excellent
7	14 Mar - 18 Mar	TestNG annotations, assertions, XML config, BDD Cucumber feature files, step definitions	Integrated TestNG with Selenium, wrote first BDD feature file with Gherkin syntax	Excellent
8	21 Mar - 25 Mar	Hackathon project - Task 1 development, Course search and data extraction automation	Automated Coursera course search with filters, extracted course data into ArrayList	Excellent
9	28 Mar - 1 Apr	Hackathon - Task 2 (Language extraction) and Task 3 (Form validation) development	Extracted language data using HashMap, captured form validation error message	Excellent
10	4 Apr - 8 Apr	DevOps - Git, Maven, Jenkins pipeline setup,	Pushed project to GitHub, configured Jenkins CI job,	Excellent

		Selenium Grid, ExtentReports integration	generated HTML ExtentReport	
11	11 Apr - 15 Apr	API automation with Postman and REST Assured, Docker basics, GenAI fundamentals	Wrote REST Assured test scripts, learned Docker containerization concepts and GenAI tools	Excellent
12	18 Apr - 22 Apr	Final project delivery, code review, documentation, Final Project and Technical Evaluation	Successfully delivered complete automation framework, cleared Final Evaluation with Excellent	Excellent

CHALLENGES AND SOLUTIONS

8.1 Challenge 1 – Dynamic Lazy Loading on Coursera

Problem: The Coursera search results page loads course cards dynamically as the user scrolls down. When the automation script tried to extract course data immediately after the page loaded, many course elements were not yet present in the DOM, causing NoSuchElementException errors.

Solution: JavascriptExecutor was used to scroll the page to specific positions before attempting to read course data. Explicit Waits using ExpectedConditions.visibilityOfElementLocated were added to pause script execution until the target course card elements were fully visible in the viewport before the getText() calls were made.

8.2 Challenge 2 – Nested Dropdown Data Extraction

Problem: Extracting the language-level data in Task 2 involved reading multiple nested filter menus. The filter items were dynamically loaded and some items were not visible without scrolling within the filter panel itself, making standard findElements() calls return incomplete lists.

Solution: A scrollable container div was identified using its CSS selector and scrolled using JavascriptExecutor. The script was structured to scroll incrementally within the filter panel and capture all language and level items using findElements() with a list-based loop, storing results in a HashMap with nested ArrayLists.

8.3 Challenge 3 – Locating the Form Below the Page Fold

Problem: The Ready to Transform form on the Courses for Campus page was positioned far below the initial viewport. Direct interaction with form fields before scrolling to the form caused ElementNotInteractableException errors.

Solution: The script used JavascriptExecutor.scrollIntoView() to scroll each form section into view before interacting with it. WebDriverWait with ExpectedConditions.elementToBeClickable was applied to all form fields to ensure they were fully interactive before any sendKeys() or click() calls were executed.

8.4 Challenge 4 – Capturing the Validation Error Message

Problem: The email validation error message in Task 3 was rendered as a dynamically injected DOM element that only appeared after the submit button was clicked. The element had no stable ID and its XPath changed based on the page state, making it difficult to locate reliably.

Solution: The error message element was located using a relative XPath based on the email input field as an anchor point. An Explicit Wait with ExpectedConditions.visibilityOfElementLocated was used to wait for the error span to appear before calling getText(). The captured text was validated against the expected message and logged in the ExtentReport.

8.5 Challenge 5 – Environment Setup Delays

Problem: The initial setup of the development environment required installing multiple software components including JDK, Eclipse IDE, Selenium Server Standalone, TestNG plugin, and Maven. Version compatibility issues between Selenium 4.15.0 and ChromeDriver caused the first few test runs to fail with SessionNotCreatedException errors.

Solution: The Selenium Manager bundled with Selenium 4.x was used to automatically manage ChromeDriver version compatibility. The Maven POM.xml was updated with the correct dependency versions, and the project was cleaned and rebuilt to resolve all classpath issues. This resolved the compatibility errors and allowed consistent script execution.

CONCLUSION

This internship at Cognizant Technology Solutions, as part of the Generation Cognizant (GenC) Quality Engineering and Assurance program, was a highly rewarding experience that successfully bridged the gap between academic knowledge and real-world industry practice. The structured 65-day learning journey covered Core Java, SQL, Functional Testing, Selenium WebDriver, BDD Cucumber, TestNG, ExtentReports, Page Object Model, Apache POI, GitHub, Maven, Jenkins, and API Automation providing a comprehensive foundation in modern software quality assurance practices.

The Hackathon project, centered on automating three user workflows on the Coursera web application, served as the practical culmination of all concepts learned during the program. Through this project, hands-on experience was gained in browser automation, dynamic element handling, dropdown data extraction using Java collections, multi-window navigation, web form interaction, validation error capture, and CI/CD integration using Jenkins and GitHub. The framework was built following industry-standard practices including the Page Object Model design pattern, BDD Cucumber for behaviour-driven test development, and ExtentReports for professional test reporting. JIRA was used throughout for sprint management, test case documentation, and defect tracking in an Agile environment.

Beyond technical skills, the internship developed important professional competencies including structured problem-solving, attention to detail, collaborative teamwork, documentation discipline, and the ability to work within a real corporate environment under the mentorship of experienced industry professionals. Regular SME connect sessions, code reviews, and project evaluations provided continuous feedback that helped refine both the quality of the automation scripts and the overall approach to software testing.

In conclusion, this internship has been a transformative experience that significantly strengthened the technical and professional foundation required for a successful career in software engineering and quality assurance. The skills, tools, frameworks, and industry practices learned and applied during this period will serve as a strong and lasting asset in all future academic and professional endeavors.